Government Engineering College, Palakkad
Department of Electronics & Communication Engineering

**ECD 415 Project Phase 1 - Final Presentation**

# Design & Layout of CMOS BandGap Reference using SKY 130 PDK

Date : 23- 10 -2026

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# Introduction

- Bandgap Reference (BGR) is a circuit used to generate a stable reference voltage.
- It is required because most circuit parameters vary with temperature and supply voltage.
- A stable reference is essential for accurate and reliable analog/mixed-signal circuits.
- BGRs are commonly used in ADCs, DACs, LDOs, and power-management circuits.
- This project focuses on the design and layout of a BGR using the SKY130 130-nm CMOS PDK.

# Design Objectives

- Design a CMOS Bandgap Reference (BGR) using SKY130 technology with a **1.8 V** supply.
- Generate a stable reference voltage of approximately **1.2 V.**
- Evaluate temperature stability and determine the temperature coefficient (TC).
- Achieve low-power operation and quantify power consumption and supply current.
- Analyze transient response of the BGR during startup and steady-state operation.
- Verify BGR performance across PVT (Process, Voltage, Temperature) variations.

# Literature Survey

| Sl. No. | Title | Author | Journal & year | Research Contributions | Research Gaps |
|---|---|---|---|---|---|
| 1 | **Design of a Bandgap Voltage Reference for an 8-bit SAR-ADC for Powerline Monitoring using SKY130 PDK** | ***Krisna M. Canonero*** | **IEEE ICICM, 2025** | • 1.2 V CMOS BGR design in 130 nm SKY130<br>• NMOS-based CTAT and PTAT generation<br>• High-gain two-stage PMOS-input op-amp | • Low PSRR → supply-noise sensitivity<br>• Line sensitivity above target (6.7%)<br>• No layout, DRC/LVS or post-layout verification<br>• Limited PSRR improvement techniques<br>• Further optimization of biasing and op-amp design needed |

| | | | | | |
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| **2** | **A Bandgap Reference(a circuit for all season)** | **Behzad Razavi** | **IEEE SOLID-STATE CIRCUITS MAGAZINE summer 2016** | • Fundamental CTAT + PTAT BGR principle<br>• Derivation of PTAT $\Delta V_{BE}$<br>• Practical CMOS BGR architectures<br>• Low-voltag e BGR design techniques<br>• Analysis of major design trade-offs | • Op-amp offset → accuracy & drift<br>• Noise → reference degradation<br>• Poor PSRR → supply sensitivity<br>• Stability → capacitive-load limitation<br>• Start-up → reliable circuit activation<br>• Low $V_{DD}$ → difficult conventional BGR operation |

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| 3 | **A Curvature-Corrected Low-Voltage Bandgap Reference** | **Gunawan et al** | **IEEE Journal of Solid-State Circuits, June 1993** | • Curvature-corrected low-voltage BGR design<br>• PTAT current generation<br>• Temperature compensation techniques<br>• Low-voltage operation at 1 V | • Limited temperature range (0–125 ˚C)<br>• Requires trimming for low temperature coefficient<br>• Implemented using a standard bipolar process<br>• Limited supply-current performance |

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| **4** | **1.2-V Supply, 100-nW, 1.09-V Bandgap and 0.7-V Supply, 52.5-nW, 0.55-V Sub Bandgap Reference Circuits for Nanowatt CMOS LSIs** | **Osaki et al** | **IEEE Journal of Solid-State Circuits, June 2013** | • Ultra-low-power BGR and sub-BGR design<br>• PTAT voltage generation<br>• Resistor-less CMOS reference architecture<br>• 1.09 V BGR and 0.55 V sub-BGR<br>• Nanowatt power consumption | • Implemented using 0.18-µm CMOS technology<br>• BGR output depends on bipolar transistor characteristics<br>• Requires careful PVT variation control<br>• Trade-offs between low supply voltage, power consumption, and reference accuracy |

| | | | | | |
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| **5** | **A ±3 ppm/°C Single-Trim Switched Capacitor Bandgap Reference for Battery Monitoring Applications** | **Hunter and Matthews** | **IEEE Transactions on Circuits and Systems - I, April 2017** | • Switched-capacitor BGR<br>• Single-temperature trimming<br>• Offset cancellation<br>• Battery monitoring applications | • Requires clocked switching circuitry<br>• Charge injection and clock feedthrough concerns<br>• Designed for a specific battery-monitoring application |

| | | | | | |
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| 6 | **Multi-Cell Battery Sensing and Protection IC With Integrated Low-Temperature-Drift Reference for Series Battery Pack Management** | **Zhu et al** | **IEEE Transactions on Circuits and Systems - I, October 2025** | • Temperature-compensated BGR<br>• Battery sensing and protection<br>• 18-cell monitoring architecture<br>• High-voltage power management | • Complex multi-functional IC design<br>• Implemented using 180nm BCD technology<br>• Designed primarily for battery management |

| | | | | | |
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| **7** | **Serial-Stacked Single-Inductor Dual-Source Triple-Output Energy Harvesting Interface With 2D-MPPT for Battery-Free IoT Devices** | **Huang et al** | **IEEE Transactions on Circuits and Systems-I, November 2025** | • Energy-harvesting PMIC<br>• Low-voltage bandgap reference<br>• Buck-boost conversion<br>• Maximum power point tracking | • Complex multi-source architecture<br>• Depends on available harvested energy<br>• Focuses on energy harvesting rather than standalone BGR design |

| | | | | | |
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| 8 | **A 1.16-V 5.8-to-13.5-ppm/°C Curvature-Compensated CMOS Bandgap Reference Circuit With a Shared Offset-Cancellation Method for Internal Amplifiers** | **Chen et al** | **IEEE Journal of Solid-State Circuits, January 2021** | • Curvature-compensated CMOS BGR<br>• Shared amplifier offset cancellation<br>• PTAT and CTAT current generation<br>• 130nm CMOS implementation | • Requires two operational amplifiers<br>• Circuit complexity increases<br>• Uses a 3.3-V supply |

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| 9 | **A High-Precision Resistor-Less CMOS Compensated Bandgap Reference Based on Successive Voltage-Step Compensation** | **Ming et al** | **IEEE Transactions on Circuits and Systems-I, December 2018** | • Resistor-less CMOS BGR<br>• Successive voltage-step compensation<br>• Higher-order temperature correction<br>• High power-supply rejection | • Implemented in 0.5-µm CMOS<br>• Requires additional compensation circuitry<br>• Quiescent current of 38 µA |

**Table 1 : Literature Survey**

# Design Specifications

| Parameter | Target Value |
|---|---|
| Supply Voltage | **1.8 V** |
| Reference Voltage | **~1.2 V** |
| Temperature Range | **–40 to 125 ˚C** |
| Power Consumption | **<100 µW** |
| Startup Time | **<10 µs** |

Table 2 : Design Specification

# Block Diagram

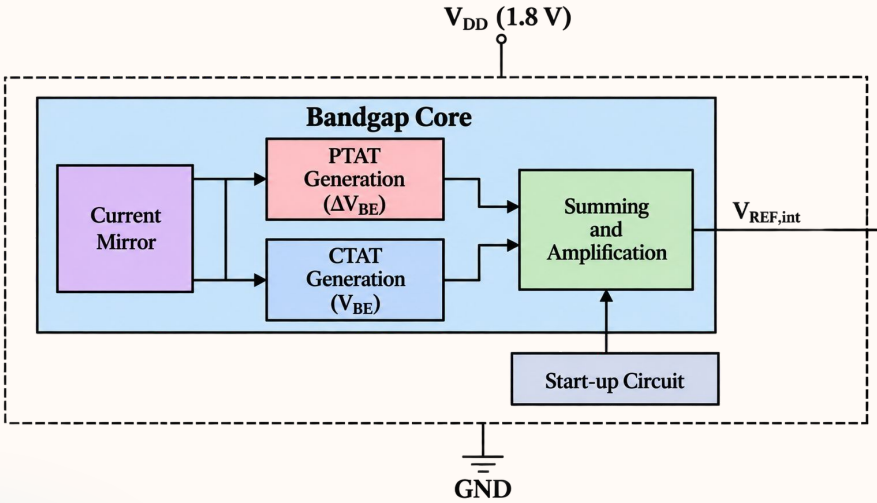



Fig 1 :Block diagram

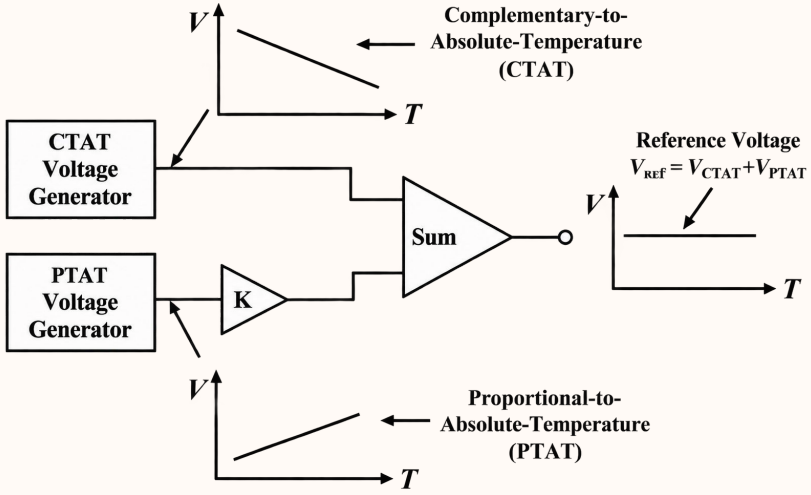



Fig 2 :Working diagram *Source(Github)*

# Working

- **CTAT**: The base–emitter voltage ($V_{BE}$) of a BJT decreases with increasing temperature.
- **PTAT:** The thermal voltage ($V_T = kT/q$) increases linearly with temperature. The difference in base–emitter voltages ($\Delta V_{BE}$) of two BJTs operating at different current densities generates a PTAT voltage.
- **Addition of PTAT and CTAT**: The PTAT voltage is scaled and added to the CTAT voltage to compensate for temperature variations.
- $$V_{REF} = V_{BE} + K(\Delta V_{BE})$$
- The resulting bandgap reference voltage remains approximately constant over temperature.

# Design Blocks

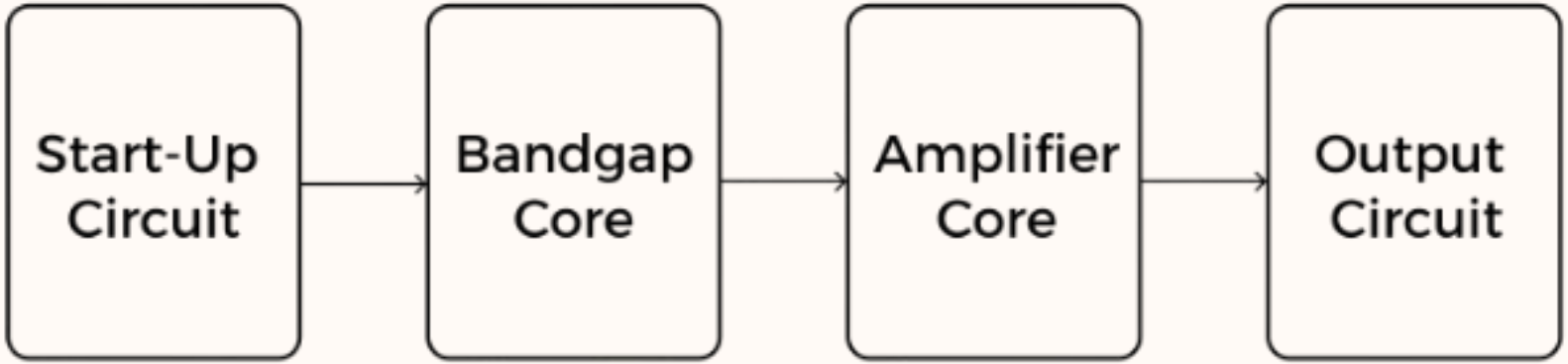



Fig 3 : Design Blocks

# Design Methodology

- **$g_m/I_D$ methodology** is a model-based technique for designing analog CMOS circuits by optimizing transistor efficiency, power, gain and speed.
- **Transconductance efficiency:**

  $g_m/I_D = g_m / I_D < 10\ V^{-1}$ : strong inversion

  $G_m$ : Transconductance
  $I_D$ : Drain Current
- **Transconductance:**

  $g_m = \partial I_D / \partial V_{GS}$
- **Key advantage:** Enables systematic transistor sizing using realistic device models instead of relying solely on square-law equations.
- **Design procedure:** Characterize MOSFETs using SPICE simulations, select the required $g_m/I_D$, determine drain current and transistor width using lookup curves, and verify performance.

# Gm/Id Simulation (Design Chart)

## NMOS Test Circuits

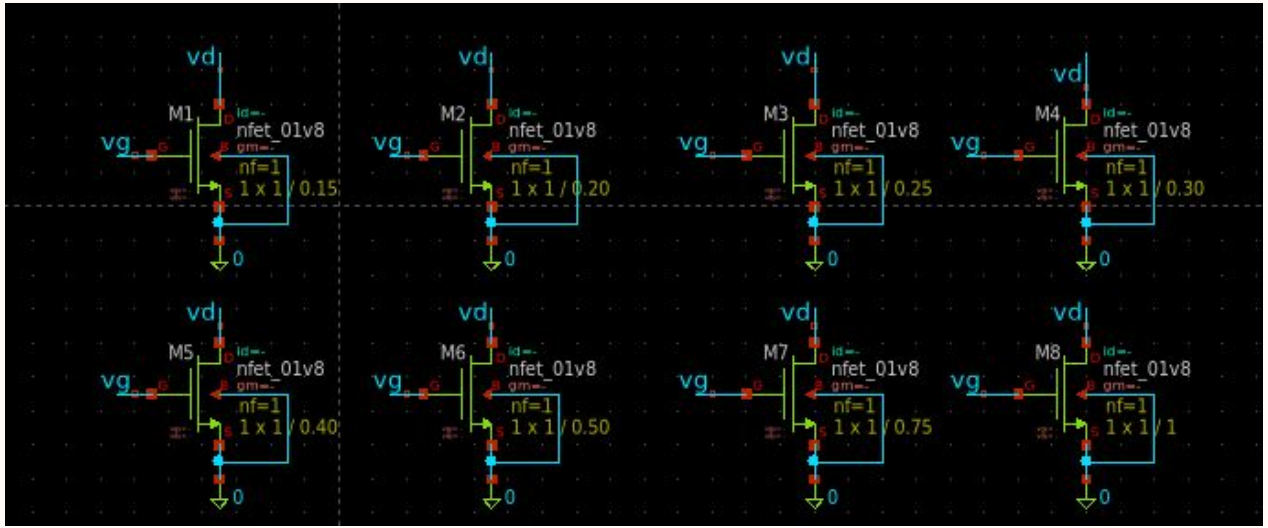



Fig 4 : Nmos test circuits

NMOS transistors with varying channel lengths (0.15–1 µm) are configured for NGSPICE simulation and parameter extraction.

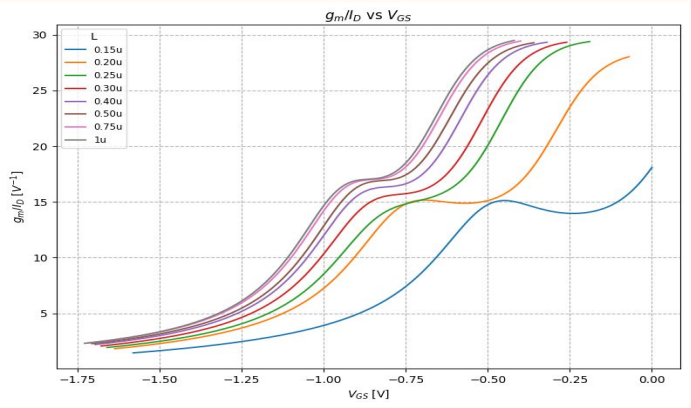



Fig 5 : NMOS $G_m/I_D$ vs. $V_{GS}$

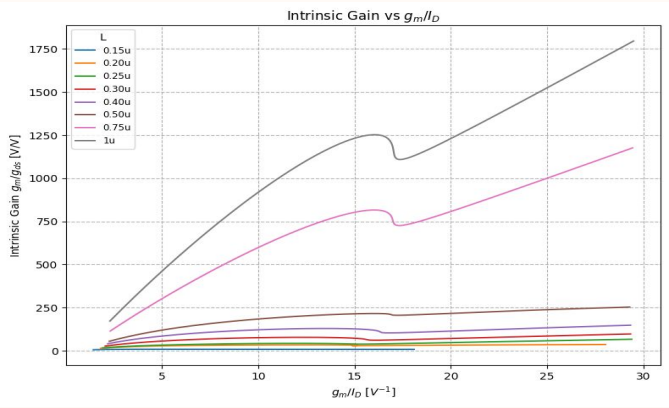



Fig 6 : NMOS Intrinsic Gain vs $G_m/I_D$

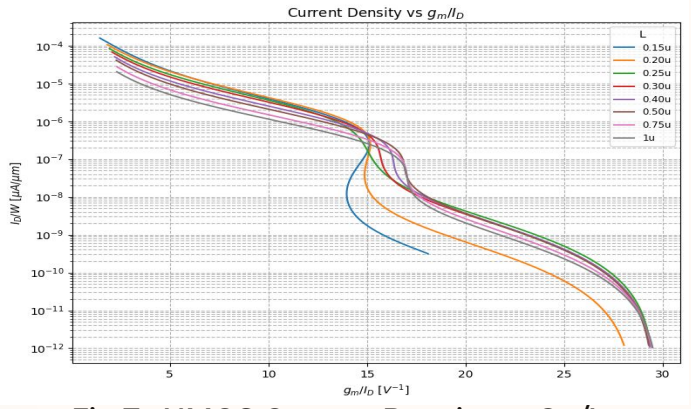



Fig 7 : NMOS Current Density vs $G_m/I_D$

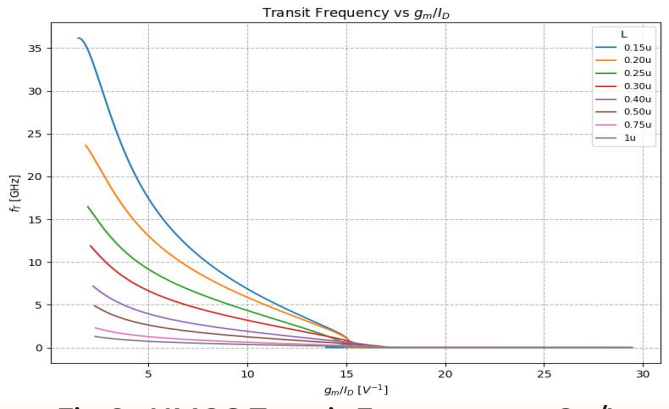



Fig 8 : NMOS Transit Frequency vs $G_m/I_D$

# PMOS Test Circuits

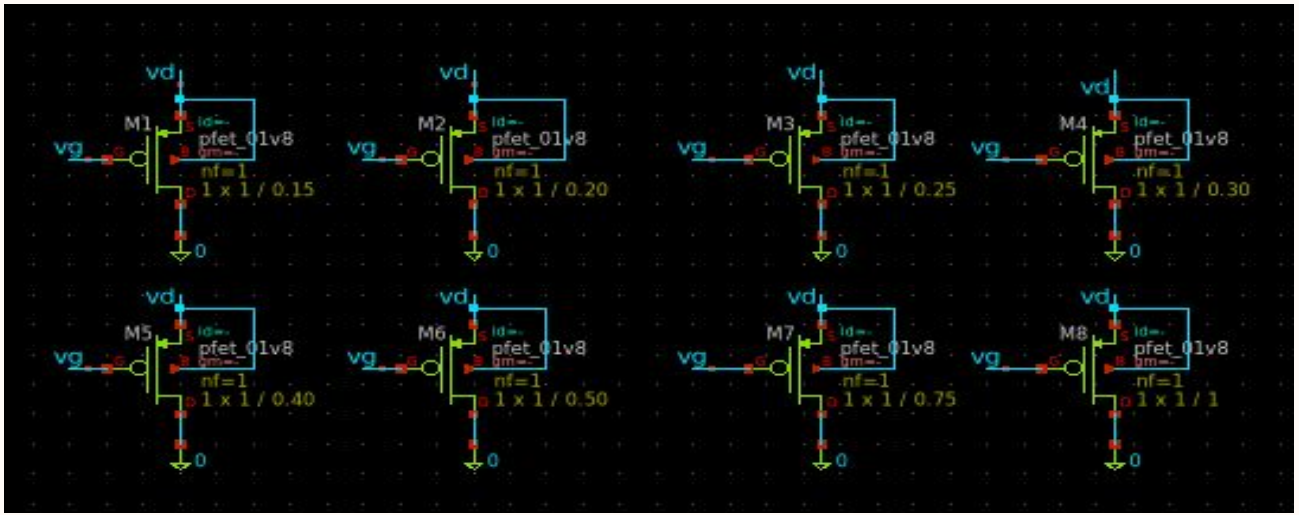



Fig 9 : Pmos test circuits

PMOS transistors with varying channel lengths (0.15–1 µm) are configured for NGSPICE simulation and parameter extraction.

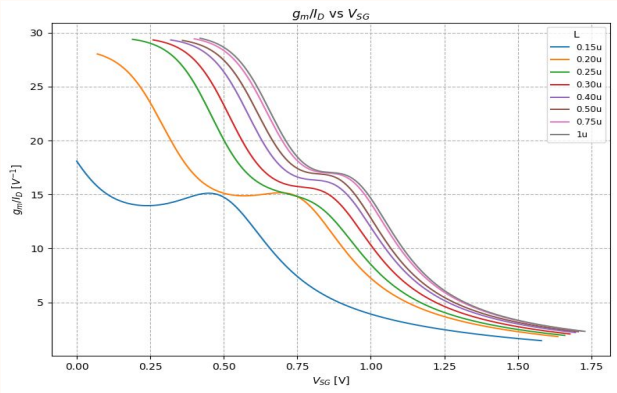



Fig 10 : PMOS $G_m / I_D$ vs. $V_{GS}$

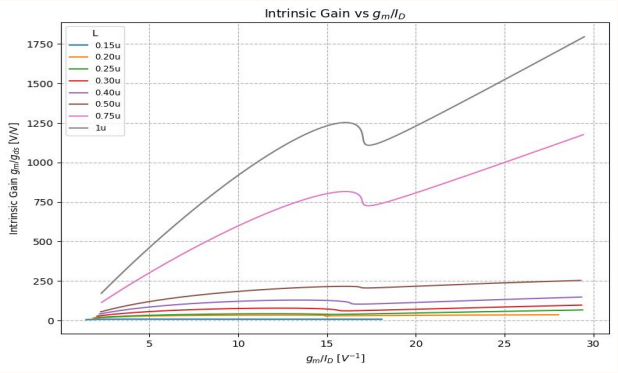



Fig 11 : PMOS Intrinsic Gain vs $G_m / I_D$

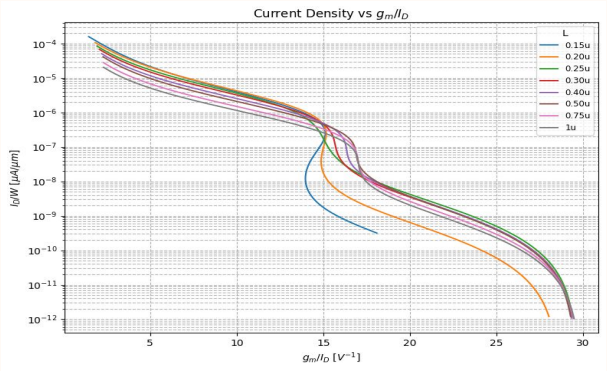



Fig 12 : PMOS Current Density vs $G_m / I_D$

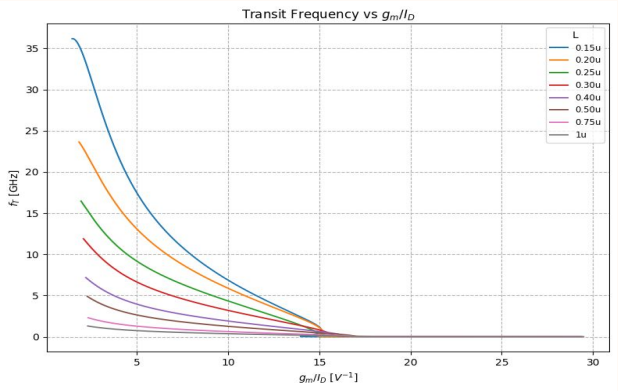



Fig 13 : PMOS Transit Frequency vs $G_m / I_D$

# Topology Selection

| Circuit Block | Topology Types |
|---|---|
| BGR Core | Basic core; PTAT current generation; Temperature-compensated core R2 & R3 |
| Amplifier | Ideal op-amp; Five-transistor OTA; Two-stage CMOS op-amp |
| Output stage | Current-output core; Voltage-output core; Regulated-cascode voltage output |
| Startup Circuit | Timing-based startup; Voltage-comparison startup |

Table 3 : Topology Selection

**Selected topology:** Temperature-compensated core + Five-transistor OTA + Voltage-output core + Voltage-comparison startup

# Startup Circuit

**Voltage-Comparison Startup Technique**

- $A_3$ (OTA/Comparator): Compares BGR output voltage ($V_{out}$) with startup reference voltage ($V_r$).
- MS (Startup Transistor): Supplies startup current from node P to activate the main PMOS transistors.
- Power-up: Vout is initially low; $A_3$ detects insufficient output voltage.
- Startup activation: $A_3$ turns ON $M_S$, helping the BGR core start.
- Normal operation: Vout rises towards its target value; $A_3$ turns OFF MS when the threshold is reached.
- Key advantage: Startup assistance is controlled by actual output voltage, avoiding unnecessary intervention during normal operation.

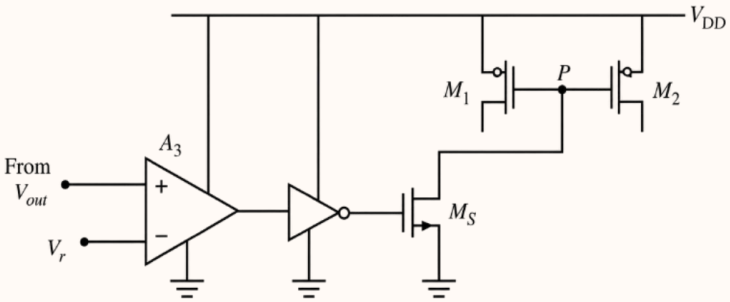



Fig 14 : Voltage comparison startup

# Bandgap Core

**CTAT Voltage**

The base–emitter voltage of a BJT is given by:

$$V_{BE} = V_T \ln(I_o/I_S),$$ $V_T = kT/q$ , $I_o$ = Bias current, $I_S$ = Saturation current

The temperature coefficient is:

$$dV_{BE}/dT = [V_{BE} - (4 + m)V_T - Eg/q]/T$$

where m ≈ −1.5 represents the temperature dependence of carrier mobility.

At T = 300 K , $V_{BE}$ ≈ 750 mV and $V_T$ ≈ 25.85 mV :

$$dV_{BE}/dT \approx -1.5 \text{ mV/°C}$$

Hence, $V_{BE}$ decreases with increasing temperature, exhibiting a CTAT characteristic.

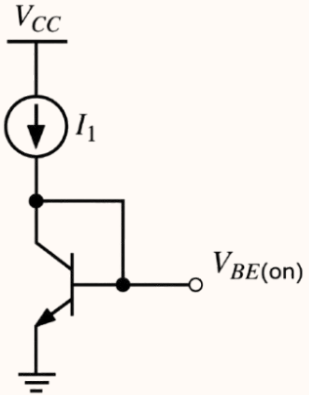



Fig 15 :CTAT generator

## PTAT Voltage

The base–emitter voltage of a BJT is given by:

$$V_{BE} = V_T \ln(I_C/I_S),$$ where $V_T = kT/q$

For two BJTs with equal collector currents and an emitter-area ratio n:

$$\Delta V_{BE} = V_{BE1} - V_{BE2} = V_T \ln(n)$$

Since $V_T = kT/q$:

$$\Delta V_{BE} = (kT/q) \ln(n)$$

The temperature coefficient is:

$$d(\Delta V_{BE})/dT = (k/q) \ln(n)$$

where $k = 1.38 \times 10^{-23}$ J/K, $q = 1.602 \times 10^{-19}$ C, and n is the emitter-area ratio.
At T = 300 K, for n = 16: $\Delta V_{BE} = 25.85$ mV × ln(16) ≈ 71.7 mV

$$d(\Delta V_{BE})/dT \approx +0.24\ mV/^{\circ}C$$

Hence, $\Delta V_{BE}$ increases linearly with temperature, exhibiting a PTAT

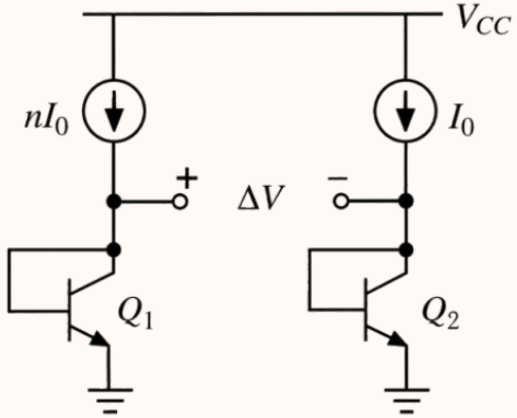



Fig 16 :PTAT generator

## Combining CTAT and PTAT for Temperature Compensation

- CTAT voltage decreases with temperature: $dV_{BE}/dT \approx -1.6$ mV/K.
- PTAT voltage increases with temperature: $\Delta V_{BE} = (kT/q) \ln(n)$.
- The weighted sum compensates for temperature variations:

$$V_{REF} = V_{BE} + K V_T \ln(n)$$

For zero first-order temperature coefficient:

$$-1.6 + K \times 0.086 \times \ln(n) = 0$$

Required scaling factor for n = 16:

$$K \ln(n) \approx 1.6/0.086 \approx 18.6$$
$$K \approx 18.6/\ln(16) \approx 6.7$$

Reference voltage at 300 K:

$$V_{REF} \approx 0.75 + 6.7 \times 71.7 \text{ mV} \approx 1.23 \text{ V}$$

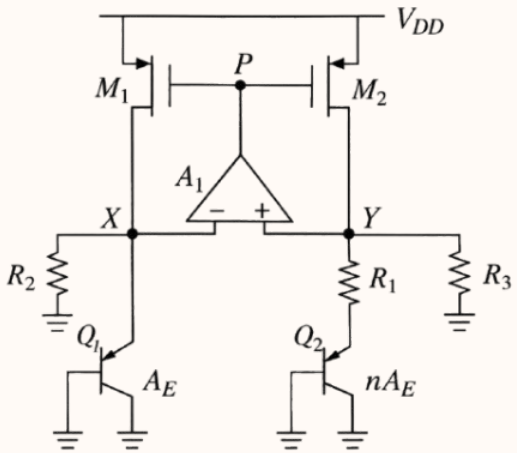



Fig 17 : BGR Core

- The scaling factor K is determined by the resistor ratio $R_2/R_1$.
- The resulting reference voltage is approximately 1.2 V, close to the silicon bandgap voltage, with a near-zero first-order temperature coefficient.

# Amplifier Topology

**Five-Transistor OTA (5T OTA)**

- NMOS Differential Pair ($M_x$, $M_\gamma$): Converts input voltage differences into current differences.
- PMOS Active Load ($M_a$, $M_\beta$): Provides current mirroring and voltage gain.
- Tail-Current Source (50 µA): Sets bias current, affecting gain, speed, and power.

Key Equations

$$I_{SS} = I_{D,}M_x + I_{D,}M_\gamma$$

For $V_x = V_\gamma$: $I_{D,}M_x \approx I_{D,}M_\gamma \approx 25\ \mu A$

$$V_{ID} = V_x - V_\gamma$$

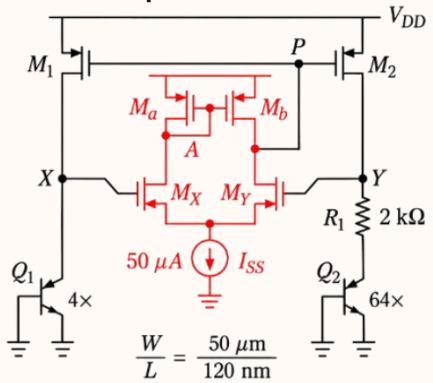



Fig 18 : 5 T OTA Topology circuit

Function in BGR: The OTA controls node P through negative feedback, equalizing the branch voltages and establishing the required operating currents.

# Output Stage

- **PMOS Transistor ($M_3$):** Copies the compensated reference current from the BGR core into the output branch using current-mirror action.
- **Load Resistor ($R_L$):** Converts the copied current into the required reference voltage.
- **Working Principle:** The compensated current flows through (M3) and ($R_L$), generating a stable output voltage.
- **Output Equation:**

  $V_{out} = I_{M3} \times R_L$
- **Design Target:** Approximately (1.2,V) reference output using a (1.8,V) supply.
- **Advantage:** Simple topology that generates a usable reference voltage with a compact output stage.

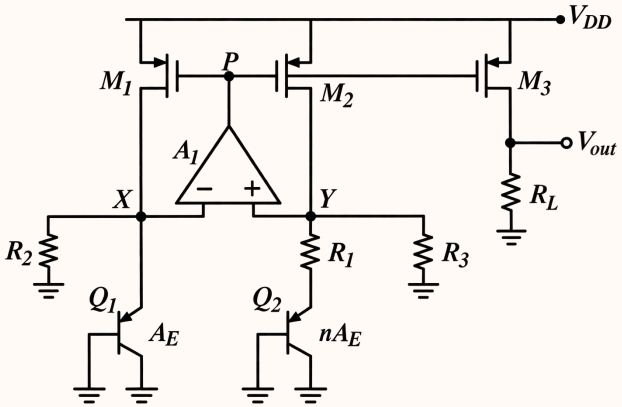



Fig 19 : Voltage core output circuit

# Design Flow

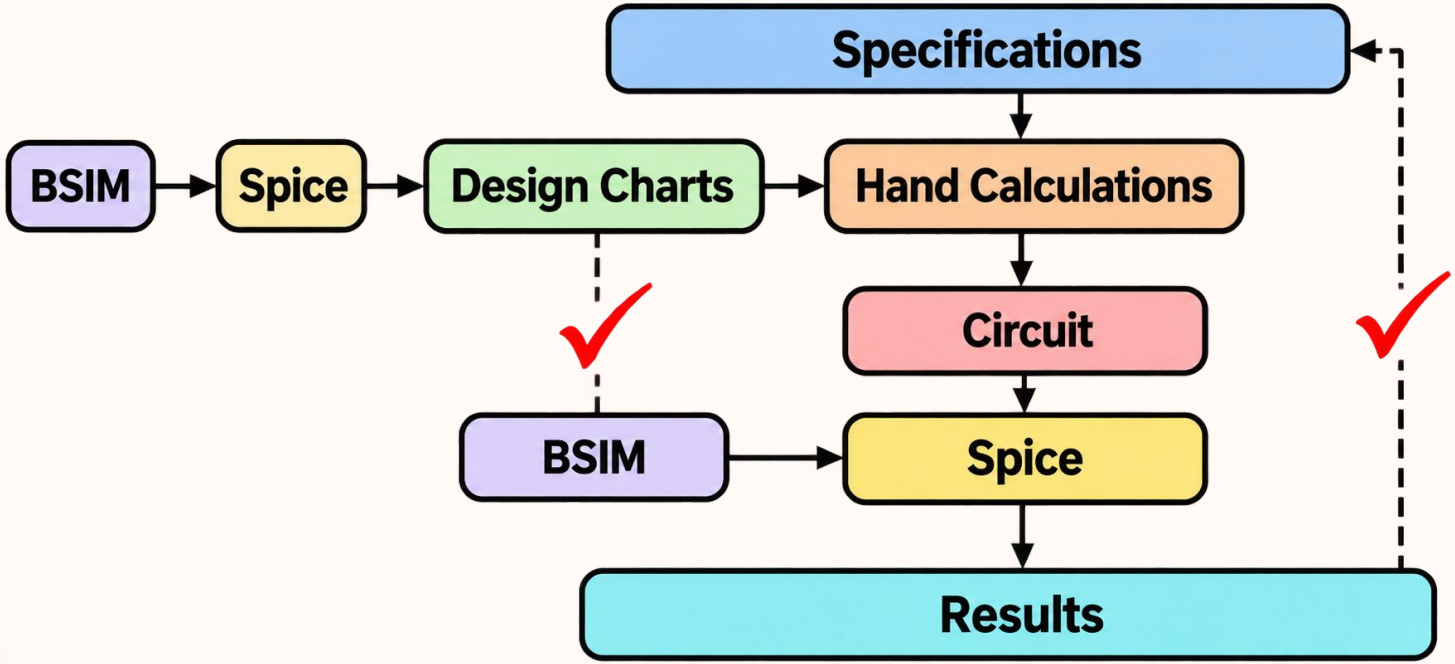



Fig 20 : Design Flow

# Tools

| Tool | Function | |
|---|---|---|
| **Xschem** | BGR schematic design, device selection, circuit connections and SPICE netlist generation. | Fig 21: Xschem *source(google)* |
| **NGSpice** | BGR simulation using SKY130 models; analyzes **Vref, temperature, supply dependence, bias current and power**; helps optimize device sizes and resistor values. | Fig 22: NGSpice *source(google)* |
| **KLayout** | BGR physical layout design, layer visualization, placement, routing and physical verification. | Fig 23: Klayout *source(google)* |

Table 4 : Tools

# Ideal Bandgap Reference Design

- Topology: Ideal BGR core with BJT pair, PMOS current mirrors, resistors, and ideal amplifier.
- PTAT–CTAT Generation: Combines $\Delta V_{BE}$ (PTAT) and $V_{BE}$ (CTAT) for temperature compensation.
- Output Stage: PMOS current mirror M3 and load resistor RL generate Vout
- Simulation Result: Approximately 1.2 V output with minimal variation across the temperature sweep.
- Emitter area ratio of BJTs is 16:1
- *Note: Startup circuitry is excluded in this idealized design.*

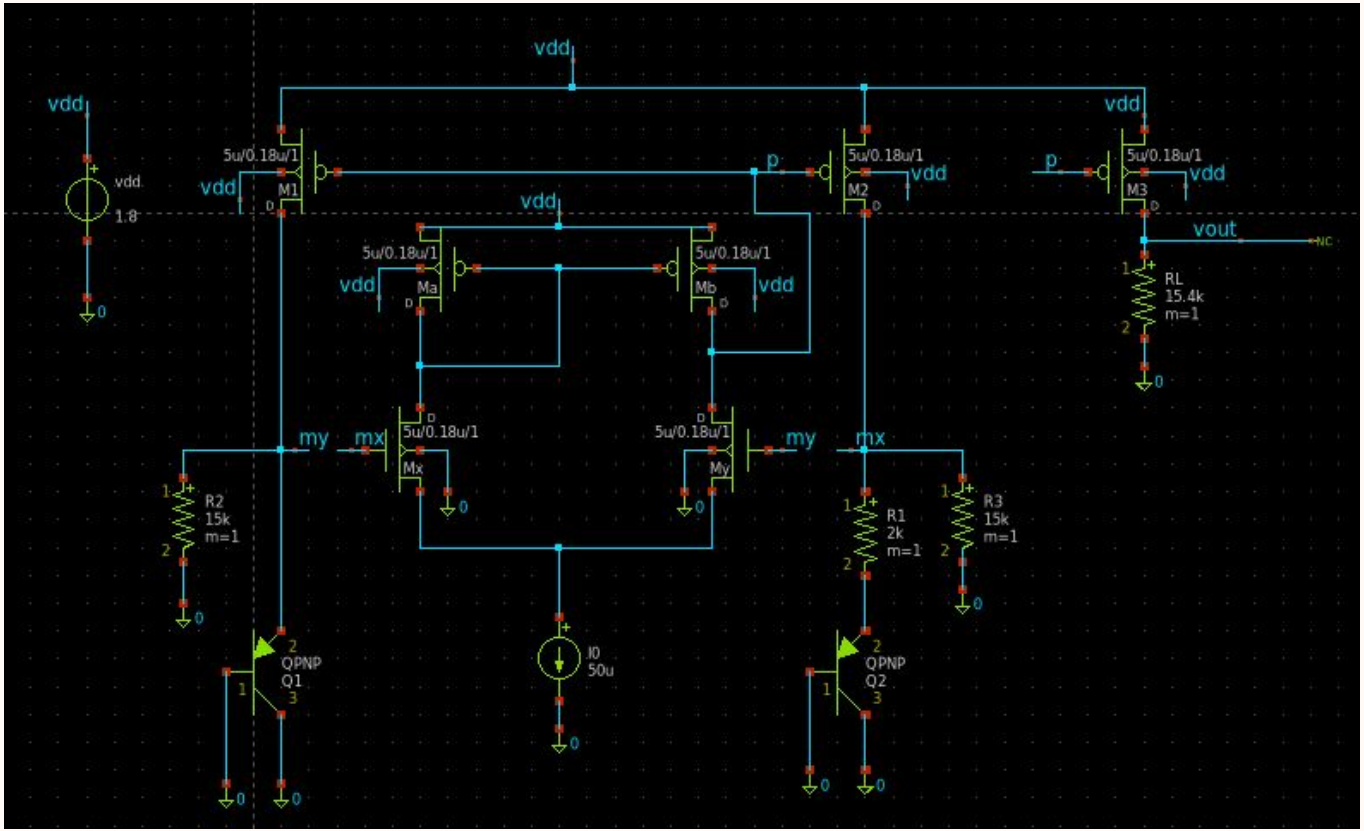



Fig 24 : Ideal BGR Circuit

| Parameter | Design Equations | Calculated Value |
|---|---|---|
| **Thermal voltage** | $V_T = KT/q$ | 25.86 mV |
| **PTAT voltage** | $\Delta V_{BE} = VTln(16)$ | 71.72 mV |
| **PTAT resistor** | $R1 = \Delta V_{BE} / IPTAT$ | 2 kΩ |
| **PTAT current** | $I_{PTAT} = \Delta V_{BE} / R_1$ | 35.86 µA |
| **CTAT voltage (assumed)** | $V_{BE}$ | 0.662 V |
| **PTAT contribution** | $V_{PTAT} = V_{REF} - V_{BE}$ | 0.538 V |
| **Scaling resistor** | $R2 = V_{PTAT} / I_{PTAT}$ | 15 kΩ |
| **Scaling Ratio** | $K = R_2 / R_1$ | 7.5 |
| **Reference voltage** | $V_{REF} = V_{BE} + K\Delta V_{BE}$ | ~1.2 V |
| **Output load** | $R_L$ | 15.4 kΩ |

Table 5 : Design

# Simulation Results

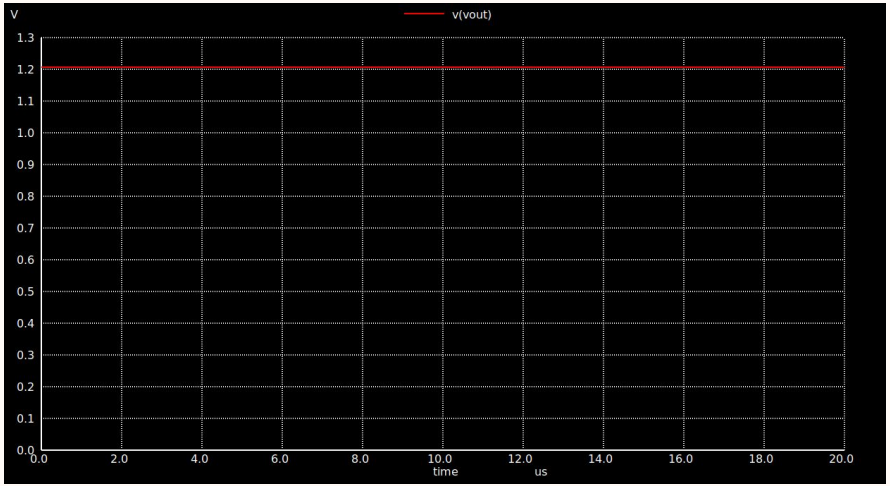



Fig 25 : Transient Response

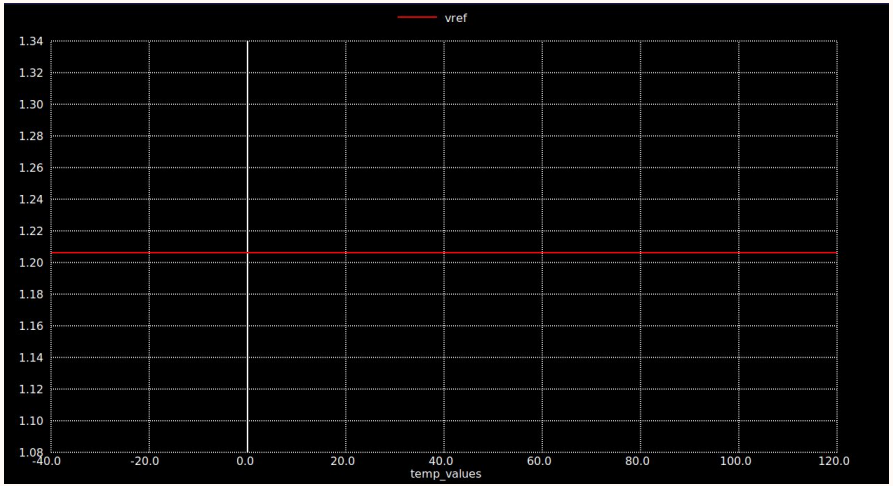



Fig 26 : Temperature Sweep

# Simulation Results

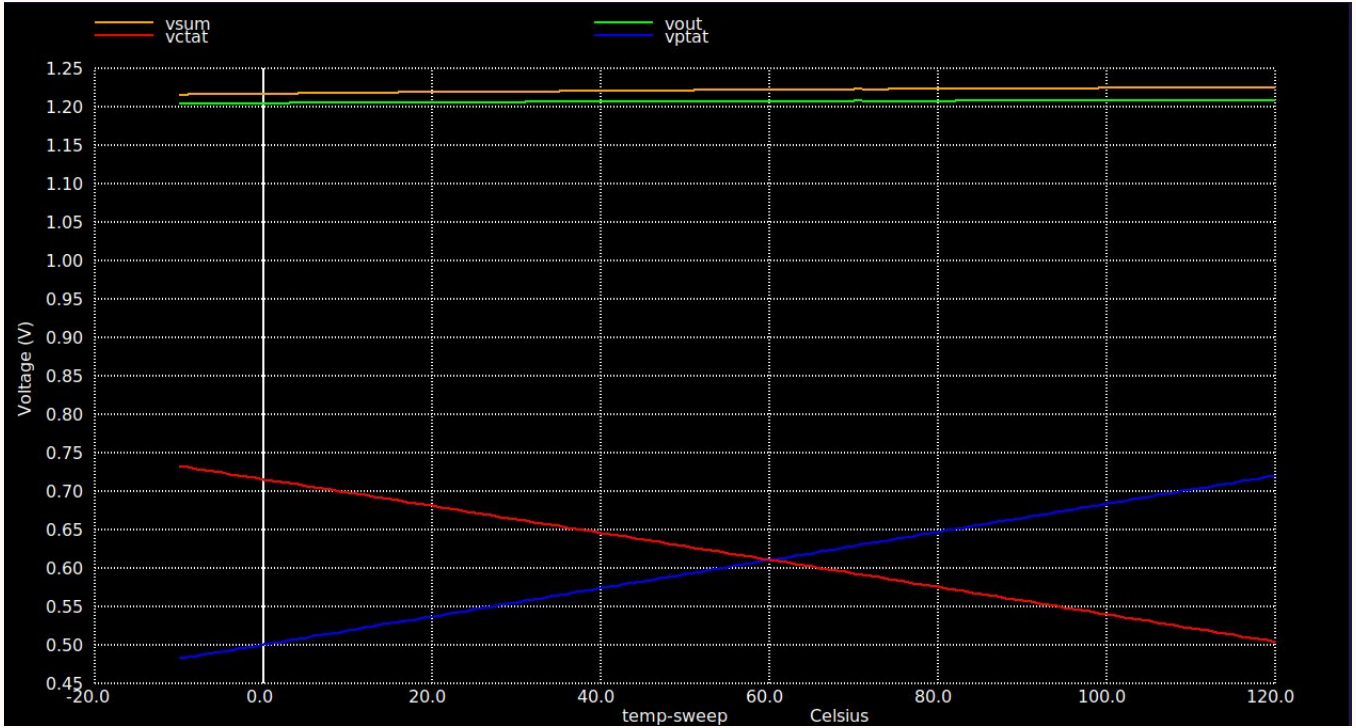



Fig 27 : PTAT–CTAT Characteristics

# Individual Contributions

| Team Member | Individual Contribution |
|---|---|
| **Aysha K H** | Literature survey and analysis of existing BGR designs |
| **Gopika P S** | BGR circuit analysis, PTAT–CTAT generation and design calculations |
| **Harikrishnan K S** | Circuit design, SKY130 PDK simulation and performance analysis |
| **Shadin P N K** | Design methodology, layout workflow, verification and documentation |

# Future Scope

- AI-Based Optimization: Use machine learning to optimize transistor dimensions and resistor values for improved BGR performance.
- Intelligent Temperature Compensation: Explore AI-assisted techniques to minimize temperature-dependent variations in the reference voltage.
- Automated Design Optimization: Integrate AI with circuit simulation tools to accelerate design exploration and improve design efficiency.
- Performance Improvement: Optimize power supply rejection ratio (PSRR) and line and load regulation.
- Explore further improvements for reliable, low-power analog IC applications

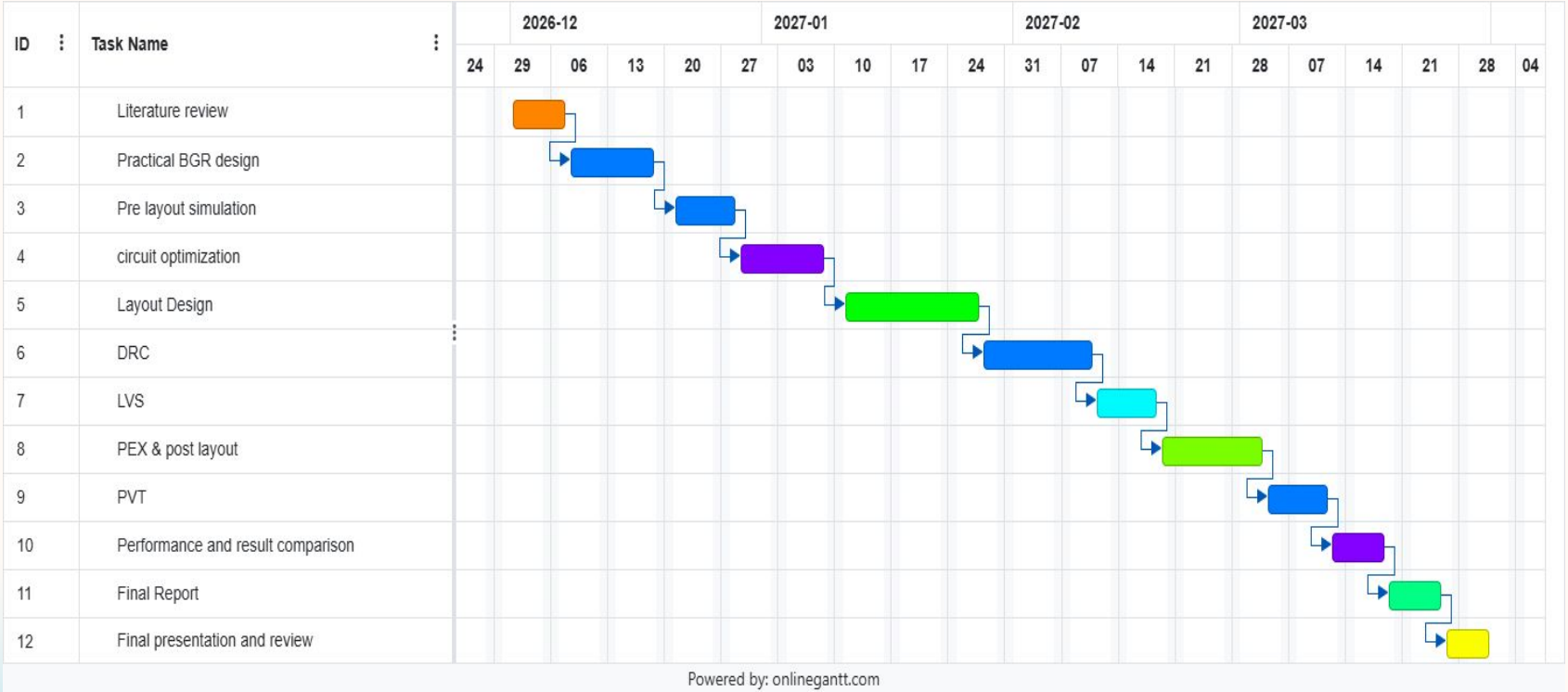
# Phase 2 - Proposed Work

- Implement the practical transistor-level BGR circuit using SKY130 PDK.
- Perform pre-layout simulations to verify reference voltage and temperature stability.
- Optimize transistor dimensions, resistor values, and bias currents.
- Design the physical layout of the BGR circuit using KLayout.
- Perform Design Rule Check (DRC) to verify layout compliance with SKY130 design rules.
- Perform Layout Versus Schematic (LVS) verification.
- Extract parasitic resistance and capacitance from the layout.
- Perform post-layout simulations to evaluate circuit performance.
- Analyze performance across process, voltage, and temperature (PVT) variations.
- Verify power consumption, startup time, and reference voltage stability.
- Prepare the final project report and presentation.

# Phase 2 -Timeline

| Month | Planned Activities |
|---|---|
| **December** | Literature Survey & Practical BGR circuit design using SKY130 PDK and pre-layout simulation |
| **January** | Circuit optimization and physical layout design using KLayout |
| **February** | DRC, LVS verification, parasitic extraction and post-layout simulation |
| **March** | PVT analysis, performance evaluation, final report preparation and presentation |

# Phase 2 - Gantt Chart



# Conclusion

- Designed an ideal CMOS Bandgap Reference (BGR) using the SKY130 130-nm CMOS PDK.
- Generated PTAT and CTAT components to achieve temperature compensation and a reference voltage of approximately 1.2 V in simulation.
- Utilized IIC-OSIC-TOOLS with open-source EDA tools for circuit design, simulation, layout, and verification.
- Established a foundation for physical implementation and post-layout performance evaluation.

# References

***Referred Papers :***


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2) *The Design of a Low-Voltage Bandgap Reference Behzad Razavi, IEEE, 2021*
3) *A Curvature-Corrected Low-Voltage Bandgap Reference — Gunawan et al. (1993)*
4) *1.2-V Supply, 100-nW, 1.09-V Bandgap and 0.7-V Supply, 52.5-nW, 0.55-V Sub Bandgap Reference Circuits for Nanowatt CMOS LSIs — Osaki et al. (2013)*
5) *A ±3 ppm/°C Single-Trim Switched Capacitor Bandgap Reference for Battery Monitoring Applications — Hunter and Matthews (2017)*
6) *Multi-Cell Battery Sensing and Protection IC With Integrated Low-Temperature-Drift Reference for Series Battery Pack Management — Zhu et al. (2025)*
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8) *A 1.16-V 5.8-to-13.5-ppm/°C Curvature-Compensated CMOS Bandgap Reference Circuit With a Shared Offset-Cancellation Method for Internal Amplifiers — Chen et al. (2021)*
9) *A High-Precision Resistor-Less CMOS Compensated Bandgap Reference Based on Successive Voltage-Step Compensation — Ming et al. (2018)*

***Textbooks :***

1. *Design of Analog CMOS Integrated Circuits, Behzad Razavi*
2. *CMOS Analog Circuit Design, Philip E Allen,Douglas R Holberg*



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# Thank You